

Active interposer technology for chiplet-based advanced 3D system architectures

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Abstract—We report the first successful technology integration of chiplets on an active silicon interposer, fully processed, packaged and tested. Benefits of chiplet-based architectures are discussed. Built up technology is presented and focused on 3D interconnects process and characterization. 3D packaging is presented up to the successful structural test and characterization of the demonstrator.

Keywords: 3D integration, active interposer, chiplet partitioning, through silicon via (TSV), Cu pillar, assembly

I. INTRODUCTION

Targeting high performance computing (HPC) applications, large many-core three-dimensional (3D) systems with multiple chiplets integrated on a silicon active interposer have been proposed [1]. The use of fine pitch 3D interconnects increases chip-to-chip bandwidth and limits overall power consumption, while a chiplet approach allows yield optimization with smaller chips and reusable IP blocks. Chiplet partitioning has already been achieved successfully and widely on passive interposers [2], but the association with active circuits on the interposer allowing smart features such as advanced network-on-chip (NoC) interconnects, fast IOs for off-chip communication, embedded power management and system-on-chip (SoC) infrastructure has not yet been reported. In this paper, we detail the first successful technology integration of chiplets on an active silicon interposer fully packaged and tested.

The paper first describes the approach of chiplet-based architectures for computing and explains its benefits in terms of memory wall breaking and cost effectiveness. Furthermore, the first part details the advantages of introducing an active circuit in the interposer, especially the capability of extended communication by using network-on-chip within the interposer. The practical case of INTACT prototype will then be presented. This object integrates a total of 96 cores offering a low power computing fabric with innovative cache-coherent architecture and wide voltage range. The 3D stack is composed of six identical multiprocessor 22 mm² chiplets, fabricated in 28 nm fully depleted silicon-on-insulator (FDSOI) technology and face-to-face stacked on a 200 mm² 65 nm CMOS active interposer by means of ultra-fine pitch Cu pillars. These Cu pillars offer large chiplet-to-chiplet communication bandwidth through the interposer, with a reduced impact on chiplet floorplan.

Section II explores the 3D technology deployed to achieve this demonstration. It focuses on the fabrication and optimization of the 3D interconnects leading to a successful integration. In particular, it details the metallization of the through silicon vias (TSV) with an investigation on diffusion barrier. Process optimizations leading to higher TSV process robustness are discussed based on a dedicated systematic defectivity study after TSV planarization. The realization of μ -pillar with ultra-fine pitch (20 μ m) on Al pads is described. Backside process including copper nail reveal, the formation of a redistribution layer (RDL), an organic passivation and the realization of solder bumps, is exposed.

The assembly of the 3D system is presented in section III. Interposer is assembled on a 10-layer ball grid array (BGA) by means of Cu/SnAg solder bumps and a mass reflow process. Chiplets of 600 μ m thick are stacked by thermo-compression before a metal lid is added on the top and balls are attached on the BGA. The full 3D stack is characterized by cross sections and X-ray tomography.

The fourth and last part of this article describes the characterization of the entire technology. The impact of the TSV process on CMOS devices in the interposer is discussed. 3D interconnects performance is assessed from parametric electrical measurements conducted on TSV Kelvin structures and RDL/TSV/back end of line (BEOL) daisy chains. Finally, the design-for-test architecture and test strategy of the INTACT system is discussed and illustrates the first successful integration of chiplets on an active interposer.

A. Chiplet on interposer concept

Targeting high performance computing applications, large many-core three-dimensional (3D) systems with multiple chips integrated on a silicon interposer have been proposed. Two main concerns are currently driving this innovation: memory wall, and system cost.

Memory wall constraint. Stacking high bandwidth memories (HBM) onto passive silicon interposer and using standardized logical and physical memory interface achieves large DRAM memory capacity (up to 8 GBytes) and large memory bandwidth (up to 256 GBytes/s) with reduced energy cost [3]. This is achieved with 55 μ m pitch μ -bumps. HBM memories are widely used in graphics processing unit (GPU) and field-programmable gate array (FPGA) devices; they may be used for other massively parallel computing architectures with large memory demand (Fig. 1).

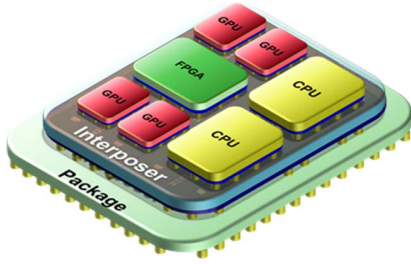


Figure 1. Chiplet on interposer topology.

Cost and form factor. Partitioning a single large die in a multitude of smaller dice leads to reduced costs in advanced nodes [4]. Stacking known-good-die (KGD) after electrical wafer sorting (EWS) reduces final system cost, as it was the case for the first FPGA partitioned in duplicated slices.

Nevertheless, there are also other qualitative reasons leading to chiplet based partitioning, heterogeneity and specialization. Instead of integrating all functions in a single CMOS node, one can use adequate technology for the right design and architecture. The case of HBM memories could be applied further, *e.g.* by integrating high-end computing cores in advanced nodes and complex analog and circuit IO physical layers (PHY) in more mature technologies. This could be achieved with a service IO chiplet offering all required system IOs. As a result, computing chiplets could be smaller and cost effective without integrating any analog and PHY.

Specialization also leads to genericity. This concept offers highly configurable and optimized chiplet, using adequate technologies: generic computing cores chiplet, GPU cores chiplet, FPGA fabrics chiplet, artificial intelligence (AI) accelerators chiplet for energy efficient execution of AI kernels. Finally, more dedicated functions can be integrated on specialized chiplets, designed and fabricated on demand, for system specialization.

The final system is then fully scalable by adding more dice or larger dice onto the interposer, while still keeping genericity, specialization and control of the system cost. Many technical challenges are still ahead, such as standardization of chiplet interfaces, system partitioning to get early system performance estimates, system level design methodologies, etc. In addition, chiplets are still currently integrated onto passive 2.5D interposers, which offer energy efficiency but rather basic chip-to-chip interfaces, and no opportunity for interposer specialization.

B. Benefits of an active interposer

By adding active circuits within the interposer, this latter becomes a complete circuit. Specialization and system optimization is then offered not only by the chiplet but also by the active interposer. Fig. 2 presents the overview of an active interposer for a computing many-core architecture.

Regarding technology partitioning and system cost, since the active interposer is assumed large, it is required to use a mature technology for the interposer to ensure high system yield, while the computing chiplets will be implemented using an advanced node.

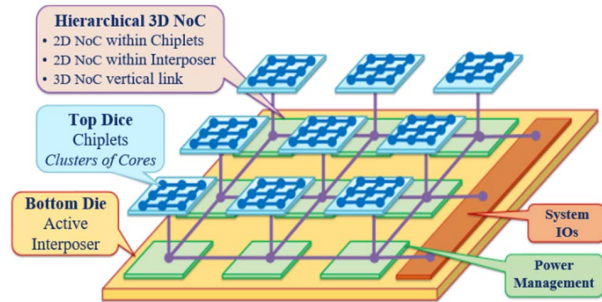


Figure 2. Active Interposer Partitioning for Many-core [1].

In terms of architecture, contrarily to passive interposer with system level communication restricted to chip-to-chip side-by-side communication, active interposers offer the possibility of extended communication capability. By using network-on-chip (NoC) within the interposer, a hierarchical NoC can be obtained, allowing communication from any chiplet to any chiplet, which is mandatory to support scalable cache coherent protocols [1][5]. With finer 3D interconnect pitch, communication bandwidth and density are increased, thus increasing the system level communication capacity.

Active interposers may also integrate system IOs as a specialization of the interposer for system level communication, and primarily with the memory.

Active interposers may also integrate power management features. The objective is to deliver the power as close as possible to computing chiplets. This allows reducing voltage drops and converter reaction time, and increases energy efficiency. Fully 3D integrated power management can be envisioned without any external passive components [6].

Regular circuit services such as clocking, reset, probes, and design-for-test (DFT) logic for testing the chiplet individually and the interposer can be easily and safely implemented within the interposer (see section IV.B).

Finally, a low logic density of 10-20 % has to be considered within the active interposer. It should be enough to provide all required and aforementioned features, while still ensuring a cost effective 3D stack [7].

C. INTACT prototype: computing with 96 cores on a chip

The INTACT prototype is a general purpose computing architecture offering a total of 96 cores (Fig. 3). An overview of its system architecture is shown in Fig. 3. The 3D stack is composed of six identical multiprocessor chiplets [8]. Each 22 mm² chiplet, fabricated in 28 nm FDSOI technology, integrates 16 cores, offering a low power computing fabric, with wide voltage range (0.6 V – 1.2 V). The chiplets are face-to-face stacked onto a 200 mm² 65 nm CMOS active interposer by means of ultra-fine pitch (20 μm) Cu μ-pillars [9]. A total of 150,000 Cu pillars are used on the active interposer to connect the six chiplets, including power-supplies. The 100 μm thick interposer contains 14,000 through silicon vias (TSV) mid-process of 10 μm in diameter. It integrates power units (switched cap DC-DC converters), system interconnects (network-on-chip), but also design-for-test, clocking and monitoring functions.

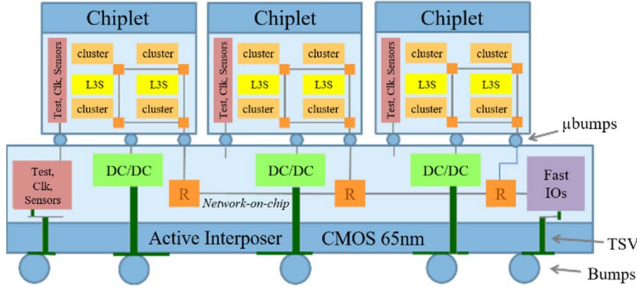


Figure 3. INTACT system architecture overview.

II. 3D TECHNOLOGY DEVELOPMENT

In this section the process technology built up for the realization of INTACT prototype is presented. Because it constitutes the vertebral column of the 3D architecture, linking chiplets, interposer and substrate, the 3D interconnects will be described in details. As already mentioned in section I, for INTACT prototype six chiplets are stacked above an active interposer by means of Cu pillar with a minimum pitch of 20 μm [10]. The 100 μm thick interposer holds TSV with a diameter of 10 μm that connect frontside BEOL to backside redistribution layer (RDL) and solder bumps. Attention will be paid to the interposer fabrication. Chiplet process will be limited to the realization of μ -bumps while BGA process will not be discussed. The interposer process flow is the following:

- CMOS 65nm front end of line (FEOL)
- TSV middle
- CMOS BEOL with 7 metal layers and Al pads
- Cu pillar above Al pads
- Temporary wafer bonding and silicon thinning
- Cu nail reveal, insulation and opening
- Redistribution layer (RDL) and passivation
- Solder bumps

A. Through silicon via development

1) TSV middle process flow

TSV middle flow is chosen for the realization of the INTACT prototype (Fig. 4). Details on this flow have been published in previous papers for 10x80 μm TSV, as in [11][12]. After front end of line (FEOL) processing up to pre-metal dielectric (PMD), a 40 nm SiN layer was deposited by chemical vapor deposition (CVD) as stop layer for future TSV planarization sequence. Lithography was done with 6 μm thick resist, with a unique diameter of 10 μm . Nitride and oxide layers were opened with dielectric etch, down to the silicon. Deep reactive ion etching (DRIE) was then used to etch the silicon with a depth of 105 μm . Etching processes were tuned to avoid silicon undercut under dielectric stack, to optimize via profile for the further metallization sequence, improving continuity of metal layer. TSV insulation was performed with the deposition of 350 nm sub-atmospheric CVD and 80 nm plasma enhanced CVD oxides. Thicknesses measured on TSV cross sections are given in Table I.

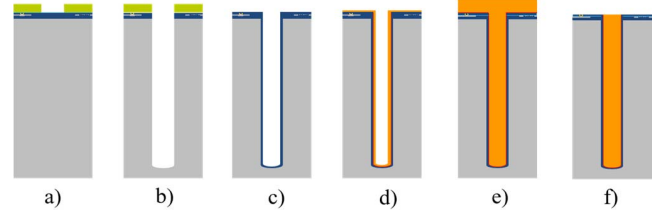


Figure 4. TSV process flow: a) lithography, b) dielectric & Si etch, c) TSV insulation, d) barrier/seed deposition, e) Cu fill, f) TSV planarization.

TABLE I. TSV INSULATION THICKNESS MEASUREMENTS

TSV region	Wafer Center	Wafer Edge
Sidewall Top	378 nm	379 nm
Sidewall Middle	330 nm	347 nm
Sidewall Bottom	319 nm	321 nm
TSV Bottom	343 nm	334 nm

After insulation, barrier and seed layers were deposited. A widely used TaN-based barrier was compared to an alternative TiN-based barrier, studied for high aspect ratio TSV integration. Among possible alternatives to TaN/Ta, that could be limited in terms of step coverage, TiN is an interesting candidate, as described in [13]. Low temperature metal organic chemical vapor deposition (MOCVD) with tetrakis(diethylamino)titanium (TDEAT) has proven high conformality with excellent barrier properties against copper diffusion [14].

For the TaN-based barrier TSV, the deposition sequence consisted in physical vapor depositions (PVD) of 10 nm TaN, 120 nm Ta and 600 nm Cu seed layers. For the TiN-based barrier TSV, the deposition sequence consisted in 20 nm MOCVD TiN, 100 nm PVD Ti and 1500 nm ionized PVD (iPVD) Cu seed layers. Conformality of the iPVD process being limited for aspect ratio higher than 10x80 μm , an additional 200 nm electro-grafted Cu layer was deposited to ensure proper seed continuity across the full TSV depth for the TiN flow. Note that the Ti cap was added on the TiN to improve the adherence of the Cu seed on the barrier.

In both cases, electro chemical deposition (ECD) of Cu was processed with an optimized recipe allowing perfect bottom-up filling of the TSV, as illustrated in Fig. 5 and Fig. 6. Consecutive to ECD Cu fill, an overburden reduction step (OBEB) was done *in situ* with diluted sulfuric peroxide (DSP) to reduce the Cu thickness on the field from $\sim 5 \mu\text{m}$ to less than 2 μm . This process helped at reducing the stress on TSV interfaces during the subsequent Cu anneal.

TSV was then annealed at 400°C with a heating profile also optimized to minimize stresses in the interfaces. Finally a three-step chemical mechanical polishing (CMP) process was performed to remove Cu, TaN or TiN barriers and SACVD oxide respectively, with selective stop on the SiN layer above PMD. At this stage the wafer surface was perfectly flat on the PMD dielectric and the remaining SiN. No adaptation was needed for TaN and TiN-based wafers as uniform stop on the SiN was achieved in both cases.

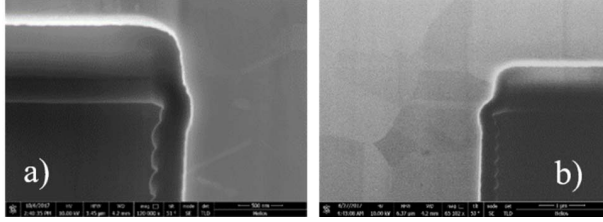


Figure 5. Cross sections of TSV with (a) TiN/Ti, (b) TaN/Ta barrier.

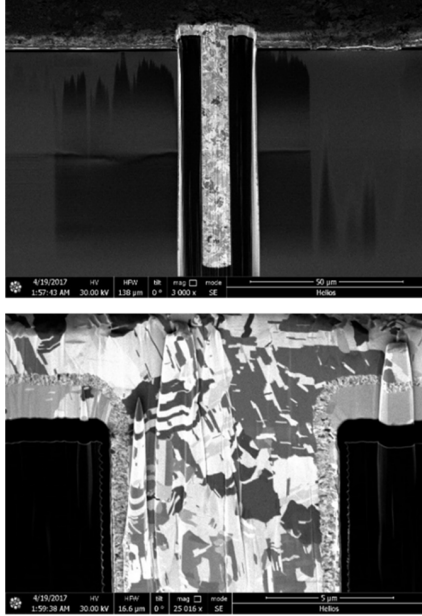


Figure 6. TSV filled with TiN/Ti barrier flow.

2) TSV defectivity management

Because wafers were re-entering the CMOS flow for BEOL processing, particular attention was paid to the control of the defectivity associated to the TSV. A systematic defectivity inspection was implemented after CMP. In addition to FEOL patterns, over 2.8 million TSVs were inspected per wafer and allows quantitative analysis. An example of data is given in Fig. 7 (a) where 938 defects are localized on a wafer. We clearly observe on Fig. 7 (b), showing all defects superposed on one field, that the main defectivity is related to the TSV, as the pattern directly corresponds to TSV layer layout. Inspection revealed that 95 to 99% of these defects, depending on the wafers, are located at the TSV surface. They have a homogeneous distribution across the wafer with a size ranging from 50 nm to 10 μm , allowing a possible classification by size, as illustrated in Fig. 7 (c). *Class 1* includes purely cosmetic defects with diameter <1 μm , *Class 2* contains defects <3 μm with no consequence for the post process and *Class 3* contains bigger defects with possible impact on BEOL process. Finally, *Class 4* (not shown) depicts large defects as cracks, layer delamination or unfilled TSV. In addition to TSV defects, random defects, estimated between 50 and 200 per wafer, are found outside TSV vicinity. They correspond to embedded particles, FEOL defects and micro-scratches.

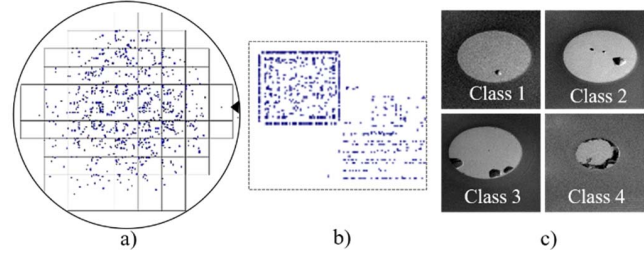


Figure 7. (a) 938 defects localized on wafer, (b) defects superposed on one field showing relation with TSV layout, (c) defect classification by size.

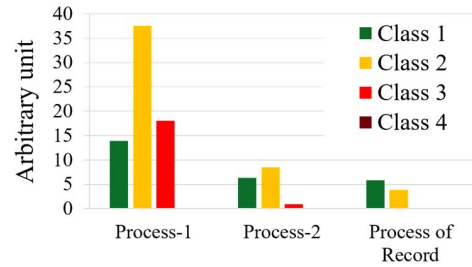


Figure 8. Defectivity analysis allowed improving overall TSV process flow quality (data from TiN-barrier wafers).

Data analysis allowed validating compatibility of wafers with the requirements of subsequent BEOL process. It also allowed improving the overall TSV flow by assessing optimizations of etch, metallization and anneal processes as shown in Fig. 8 with the distribution of defects for three evolutions of the technology. Final process of record (POR) gives lowest defect density with only *Class 1* and 2 defects.

Wafers with TiN barrier exhibit a higher defect count than TaN barrier, with a ratio reaching 5:1 on some wafers, but with lower criticality, essentially *Class 1* and 2 defects. Wafers with TaN barrier however exhibit, a larger amount of *Class 3* defects with several *Class 4* that were not detected on TiN wafers. Localized cracks and layer delaminations in relation with TSV localization were observed at the edge of two wafers with TaN barrier and caused wafers rejection. This difference seems to be related to a higher stress in the TSV region due to the TaN/Ta barrier. Full field X-Ray diffraction conducted at the European Synchrotron Radiation Facility (ESRF) confirmed this assumption. TiN-based flow may suffer from micro-voids at Ti/Cu interface due to a weakness of iPVD Cu process, leading to *Class 1* and 2 defects after anneal. This issue is not met with PVD Cu in TaN-based flow.

B. Interposer/Chipselets Interconnect Technology

1) μ -pillars on interposer

μ -pillars were processed on Al pads on interposer wafers. TiW barrier and Cu seed layers of 130 nm and 200 nm, respectively, were deposited by PVD. Lithography was achieved with 12 μm thick positive resist, with a critical dimension (CD) of 10 μm and a pitch of 20 μm . Considering this minimal pitch, a stepper process was preferred to mask aligner to avoid misalignment on Al pads and further stacking issues with the chiplets.

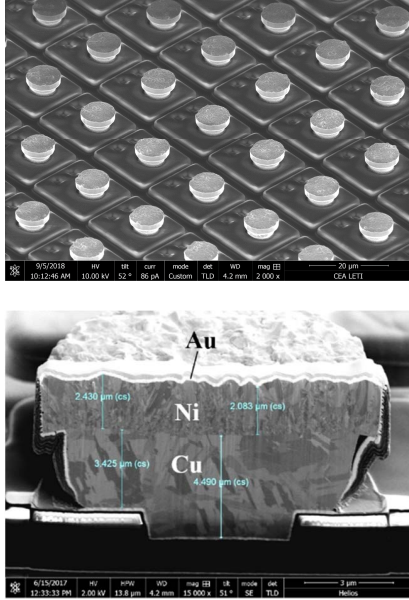


Figure 9. (top) μ -pillars array on interposer, (bottom) cross section of a μ -pillar on Al pad, composed of Cu 3 μ m / Ni 2 μ m / Au 0.1 μ m.

A descum was done before ECD step which consists in Cu 3 μ m / Ni 2 μ m / Au 0.1 μ m growth. After resist stripping, μ -pillars heights was measured by confocal interferometry on 78 points per wafer (see Table II for typical result). Cu seed and TiW barrier were etched by diluted sulfuric peroxide (DSP) and H_2O_2 , respectively. μ -pillars after full process are illustrated in Fig. 9 in an array (top), and in cross section (bottom). Reduction of Cu diameter under Ni occurred during Cu seed etching step.

2) μ -bumps on Chiplets

μ -bumps are formed above Al pads on chiplet wafers, with a process similar to μ -Pillars. ECD here consisted in Cu 3 μ m / Ni 2 μ m / SnAg 5 μ m growth. After resist stripping, μ -bumps were measured (see Table II). Cu seed and TiW barrier were etched with diluted phosphoric peroxide (DPP) and H_2O_2 , respectively. A reflow process was conducted at 255°C under N_2 atmosphere to obtain the semi-spherical shape illustrated in Fig. 10.

C. Interposer/BGA interconnect technology

1) Si thinning and Cu nail reveal

Temporary wafer bonding was achieved with 20 μ m thick adhesive and controlled by acoustic microscopy. Thinning was achieved by a combination of coarse grinding, edge trimming, fine grinding and CMP, with the objective of final Si thickness of 112 μ m. Si thickness was monitored by infrared interferometry with 4000 points per wafer. Average variation from the target was inferior to 2 μ m and typical standard deviation was inferior to 1 % within a wafer.

After proper cleaning, Cu nails were revealed with a SF_6 etch process with end point detection. Nails height ranged from 2 to 5 μ m after etching. A 2 μ m TEOS oxide was deposited at 150°C. Finally CMP was done to open the Cu for contact, with an oxide consumption of less than 300 nm.

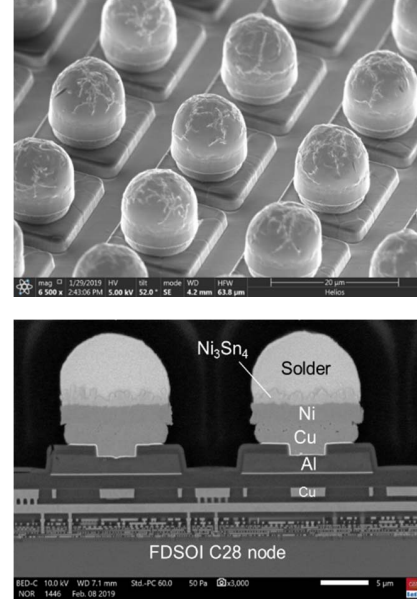


Figure 10. (top) μ -bumps array on chiplet, (bottom) cross section of a μ -bump composed of Cu 2 μ m / Ni 2 μ m / SnAg 5 μ m after reflow at 255°C.

2) Redistribution layer and passivation

Before redistribution layer (RDL) process, Cu was slightly recessed to ease alignment of RDL lithography. 100 nm Ti barrier and 200 nm Cu seed layers were deposited. Lithography was done with a 12 μ m thick resist, with minimum line width of 10 μ m (Fig. 11). After descum 2.5 μ m Cu were grown by ECD. RDL height was measured by confocal interferometry (see Table II for typical result) before Cu/Ti etching. ALX passivation was patterned on RDL with a mask aligner lithography, and annealed at 190°C for 2 h under N_2 atmosphere. Passivation openings with 55 μ m diameters defined locations of the solder bumps.

TABLE II. 3D INTERCONNECTS TYPICAL THICKNESS MEASUREMENTS (78 POINTS PER WAFER)

Interconnect	Thickness	Std. dev
μ -pillar (interposer)	5.52 μ m	0.22 μ m (3.9%)
μ -bump (chiplet)	9.61 μ m	0.47 μ m (4.9%)
RDL (interposer)	2.37 μ m	0.08 μ m (3.3%)

3) Backside solder bumps process

Solder bumps lithography was performed with a 100 μ m thick negative dry film laminated under vacuum after deposition of 100 nm Ti barrier and 300 nm Cu seed layers. Lithography was done by mask aligner with a CD of 78 μ m. Bumps were grown by ECD of 4 μ m Cu and 68 μ m SnAg. Stripping was carefully optimized for the thick film. Cu/Ti layers were etched and wafers underwent a reflow process to obtain a quasi-spherical shape with ~80 μ m in diameter, as shown in Fig. 12.

Wafers were then debonded and cleaned to remove adhesive residues; thin interposers with both sides processed were obtained.

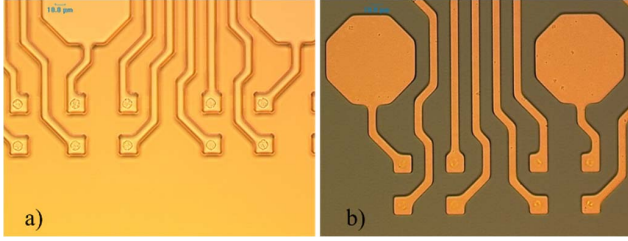


Figure 11. RDL with minimal CD of 10 μm . (a) after lithography, recessed Cu nails are visible. (b) after ECD, stripping and seed/barrier etching.

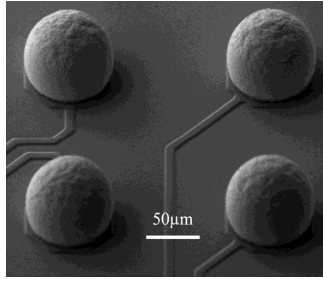


Figure 12. Solder bumps after seed etch.

III. 3D SYSTEM PACKAGING

This section details the packaging sequence used for INTACT demonstrator illustrated in Fig. 13. The interposer was first assembled on a laminate substrate by standard mass reflow (MR) process and capillary underfill (CUF). Then six 600 μm thin chiplets were stacked on the interposer frontside by thermocompression (TC) bonding followed by CUF process. Highlight is done on chiplets stacking with 20 μm pitch μ -bumps. In this process, an accurate alignment with optimization of the parallelism was needed for each chiplet.

A. Substrate description

The laminate substrate is a 40 x 40 mm² 10-layer flip chip thermal enhanced ball grid array (FCTEBGA) with 1521 backside interconnects using 1 mm pitch. Its total thickness is 1.27 mm without balls. Capacitances were mounted on the substrate before interposer assembly.

B. Interposer stacking on substrate

Prior to assembly, substrate and interposer warpages were measured as a function of temperature. Whereas substrate warpage was found constant with temperature, with an amplitude of 5 μm in interposer area and 20 μm on the whole diagonal, interposer warpage was concave with a maximal value of ~ 70 μm at 30°C (Fig. 14), and 40 μm at 230°C. Successful assembly was achieved by pick and place (P&P) and mass reflow. Hence, 80 μm diameter bumps on interposer and 15 μm solder-on-pad on substrate overcame ~ 70 μm warpage at 30°C and ~ 40 μm warpage at 230°C. After interposer assembly, X-ray microscopy (XRM) characterization was carried out, typical observation is given in Fig. 15. Attention was paid to track defects such as open and short circuits in solder bumps connections. Analysis only revealed a few defects while major part of assembled interposers exhibited perfect connections.

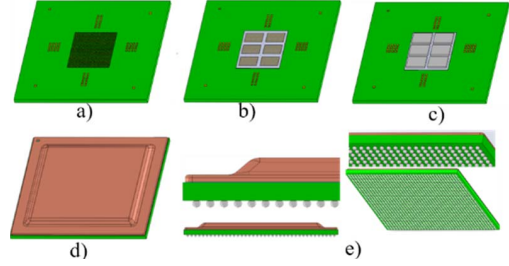


Figure 13. INTACT packaging flow: a) surface mounted devices (SMD), b) interposer assembly, c) chiplet assembly, d) lid bonding, e) balling.

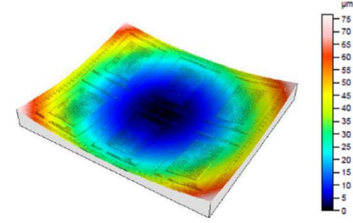


Figure 14. Interposer warpage at room temperature.

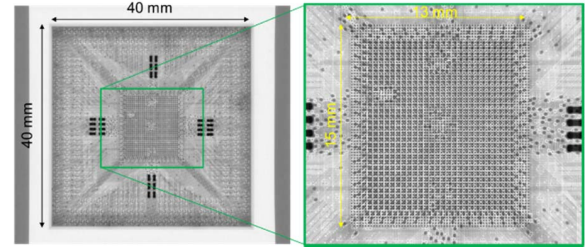


Figure 15. XRM view of the interposer on BGA, zoom on bumps location.

C. Chiplet stacking on interposer

Warpage of interposer assembled on substrate was measured as a function of temperature. In chiplet locations the warpage at 30°C was found negligible and increased to 15 μm at 230°C with a concave shape. Contact and soldering at 230°C with only 5 μm solder available on μ -bumps was then challenging. Thus thermo-compression was recommended. Successful chiplet assembly was achieved by TC on a SET FC300 tool, enabling efficient and repeatable alignment performance. This tool also enabled parallelism tuning on each chiplet. Liquid flux was used to enable solder wetting and improve the soldering during TC. Parallelism was tuned using chiplets corners. Cross sections were performed at the edge and in the center on all chiplet to check alignment performance and observe contact between chiplets and interposer. Proper connection and alignment was demonstrated on all chiplets (Fig. 16).

D. Full assembly characterization

SEM cross-sections were carried out to image all 3D interconnects levels from BGA to the chiplets (Fig. 16). X-Ray tomography also revealed the internal structure of the 3D stack with a clear view of 3D interconnects (Fig. 17). Both techniques confirmed perfect alignment and contact of μ -pillars and μ -bumps during chiplet stacking.

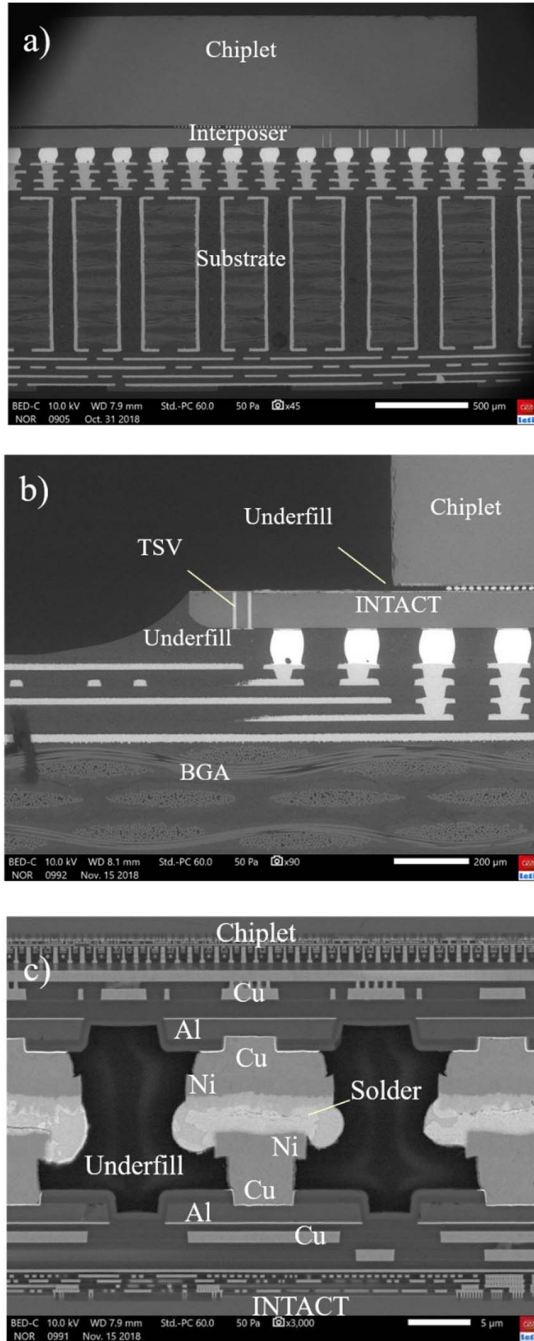


Figure 16. INTACT prototype after 3D assembly. (a) 600 μm thick chiplet on 100 μm thin interposer on BGA, (b) focus on the interposer with TSV, (c) 20 μm pitch / 10 μm diameter μ -pillars/ μ -bumps connections.

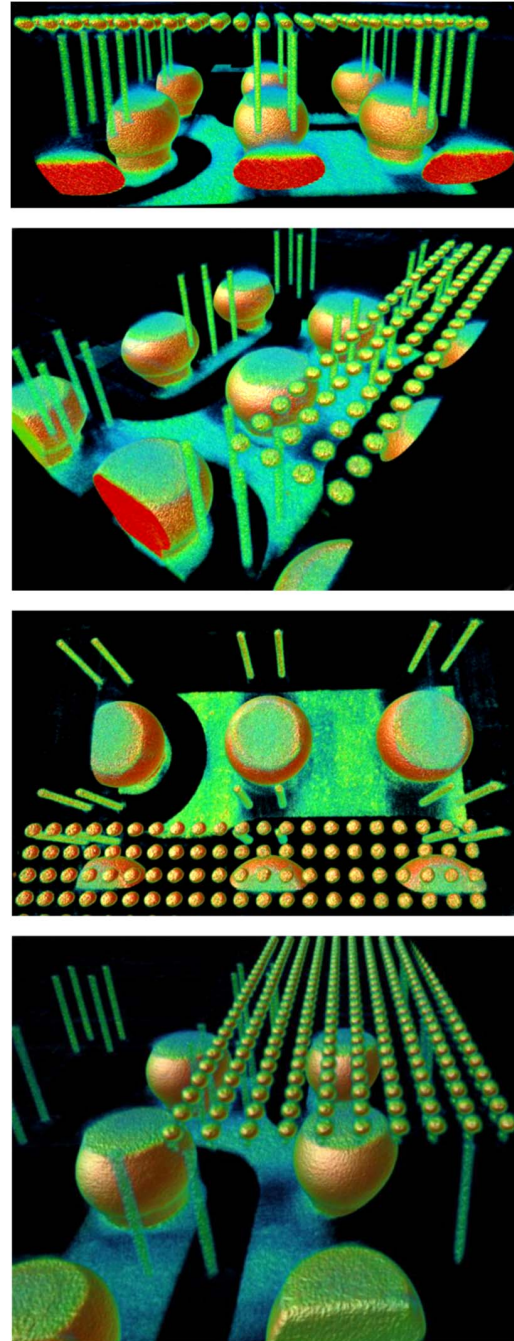


Figure 17. INTACT full 3D stack after assembly seen through X-ray tomography where all 3D interconnects layers are simultaneously visible. Perfect alignment and contact of μ -pillars and μ -bumps is observed.

IV. CHARACTERIZATION AND TEST

A. 3D process technology assessment

1) TSV impact on circuit

Parametric Test (PT) was conducted on interposer wafers during BEOL processing. TSV impact on circuit features such as MOS devices, SRAM cells, contact chains, metal layers, MIM parasitic capacitance etc. was evaluated and compared to reference lots without TSV. Fig. 18 illustrates this comparison with I_{OFF} and I_{ON} currents measured on CMOS transistors. No detrimental effect of TSV was noted during PT. However, slight differences were still observed between TaN and TiN barriers, for which we suspect a mechanical origin, explained by a lower stress level for TiN barriers.

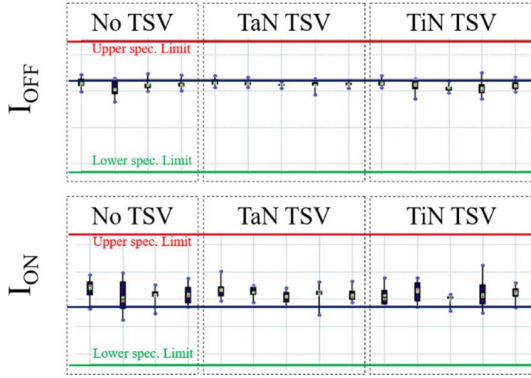


Figure 18. I_{OFF} & I_{ON} currents of MOS transistors tested at PT. TSV wafers with either TaN or TiN show equivalent behaviour than reference wafers.

2) 3D interconnects electrical characterization

This part discusses electrical characterizations of 3D interconnects conducted on RDL pads after passivation. Test structures include TSV Kelvin and RDL/TSV/BEOL daisy chains. The test was run on 75 dice per wafer, for nine wafers within three batches: batch A and batch B were processed with TaN barrier and batch C with TiN barrier.

TSV resistance was measured on Kelvin structures, avoiding resistive contributions from contacts, RDL and BEOL. Five Kelvin structures were tested on each die. Measurement results are given in Table III for the nine wafers. The unitary TSV resistance is found around 26.5 mΩ. Theoretical value for a perfect cylinder, considering 100 μm high TSV, Cu effective diameter of 9.2 μm and copper resistivity of 18 mΩ.μm, leads to 27 mΩ. These results indicate well continuous TSV without filling voids for all wafers. A limited dispersion is observed on the values within wafers and between wafers, with a maximal variation between wafers inferior to 2 %. Dispersion on TSV resistances mostly corresponds to silicon thickness variation, which is coherent with measurements conducted after wafer silicon thinning. Fig. 19 illustrates the results obtained on four wafers from batch A and B in the form of cumulative frequency graph. Measurement yield reaches 100 % for three wafers, while 7 % defective structures on wafer B1 were linked to a process issue on wafer edge.

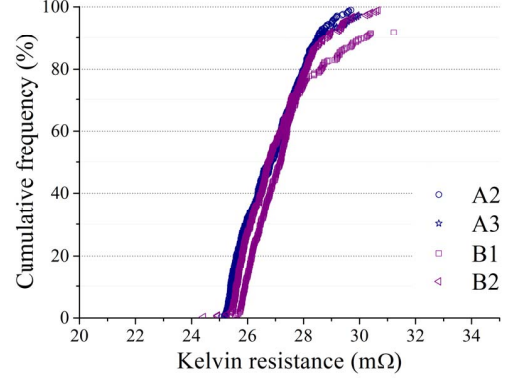


Figure 19. Kelvin resistance measured on 375 structures per wafer for A2, A3, B1 and B2 wafers.

TABLE III. RESISTANCE MEDIAN VALUE FOR THE TEN WAFERS

		Kelvin	DC 300	DC 870
Wafer	Barrier	mΩ	mΩ / TSV	mΩ / TSV
A1	TaN	26.2	169.1	169.2
A2	TaN	26.4	170.1	170.1
A3	TaN	26.4	169.9	169.8
B1	TaN	26.9	168.9	170.0
B2	TaN	26.4	169.6	169.5
C1	TiN	26.2	174.2	170.8
C2	TiN	25.7	171.9	171.4
C3	TiN	26.0	169.6	171.1
C4	TiN	26.7	173.6	172.7

Daisy chains (DC) of 30, 50, 150 and 870 TSV with 40 μm pitch were tested in 4-probe configuration to characterize the electrical yield of RDL/TSV/BEOL path. Measurement results are given in Table III for the nine wafers, and in Fig. 20 where the results on the two best wafers are illustrated in the form of cumulative frequency graph exhibiting 100 % yield on all TSV chain. Maximal variation of the resistance between wafers is about 2 % with a median value around 169 mΩ per TSV node, including RDL and BEOL resistances. A good linearity is obtained over the four chain lengths. A slight increase of the chain resistance is observed for wafers of batch C, and is explained by a thinner RDL. The higher resistance for 300 TSV DC compared to 870 TSV DC for wafers C1 and C4 indicates that some nodes in the chain are more resistive (their relative contribution in the resistance per node is lower for a long chain as DC 870). We attribute this increase to a non-optimal contact between RDL and Cu nail.

No noticeable difference was observed in TSV or chain resistance between TaN/Ta and TiN/Ti integrations, which indicates that from an electrical point of view the barrier is of second order. The characterization yield, *i.e.* the amount of functional structures, however, is lower for TiN wafers, which indicates that the integration is still not as mature as with TaN. As for the defectivity analysis, we attribute these failures to the iPVD Cu layer which presents continuity weakness. Nevertheless, the overall results exhibit high performance for full technology from FEOL to RDL.

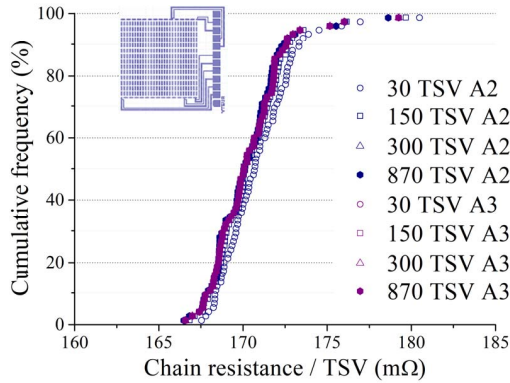


Figure 20. 30/150/300/870 TSV daisy chains measured for the two best wafers A2 & A3 (resistance is given by TSV node).

B. Active interposer demonstrator test

For such chiplet based 3D architecture, testability is a clear challenge. With the standard test challenges but also with new defect sources (TSV cracks, μ -bump misalignment, etc), a known good die (KGD) test strategy has to be defined [15]. The 3D test architecture has to offer pre-bond test of the chiplets at wafer level, before 3D assembly, and the final test after 3D assembly, while testing all features to ensure coverage of logic, memories, and 3D interconnects. Another challenge concerns the test access in the case of ultra-fine pitch 3D interconnects.

1) 3D Test Architecture

For testing such large systems with high level of maturity, a complete 3D design-for-test architecture is proposed within the active interposer [16]. It is based on two main structural test structures. The first is a JTAG test interface, using IEEE1687 standard, which offers access to boundary scan chains for individually testing all 3D interconnections (passive or active links), and for controlling memories built-in-self-test (BIST). The second is a full scan network using test compression, to test full logic stack, with a reduced pin count at 3D interface. The main design innovation is coming from chiplet footprint proposal, allowing to individually test each chiplet, in standalone for pre-bond test or mounted on the active interposer for final test, as presented in Fig. 21.

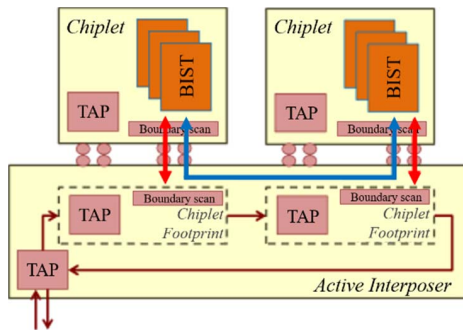


Figure 21. 3D design-for-test architecture principle. TAP stands for test access ports.

2) Test strategy

Specific care has to be paid to provide required test access structures, namely test pads for EWS probe card. Since 20 μ m pitch μ -bumps cannot be directly accessed by any standard available probe card, test is performed on a dedicated standard IO ring, integrating test signals and required power supplies.

For INTACT demonstrator, the FDSOI 28nm chiplet IO ring presents a 36 μ m pitch, which is compatible with advanced probe card cantilever technology. Fig. 22 presents chiplet IO ring where EWS test is performed, and corresponding chiplet probe card for wafer level testing.

Since the active interposer is using a mature 65 nm technology and a low logic density, yield is assumed to be very large and no test is done in pre-bond phase at wafer level (however test architecture and IO test pads are still available). After 3D assembly, final test of INTACT demonstrator is done in package using dedicated probe card and socket as shown in Fig. 23.

3) Test results

The FDSOI 28nm chiplets were successfully tested at wafer level. Binning strategy was defined according to test results (Fig. 24). For final test, the test vectors have been successfully applied to the demonstrator using an automated test equipment (ATE). It has been possible to test individually: i) the active interposer and its connection to the package, ii) the chiplet connectivity, either for chiplet-to-interposer links or for chiplet-to-chiplet links.

Correct 3D connection was achieved between the package and the interposer, including package, RDL and TSV, and between the interposer and the chiplets through μ -bumps and μ -pillars. More advanced tests are still on-going onto the ATE for full chip qualification, and for INTACT demonstration onto an application board.

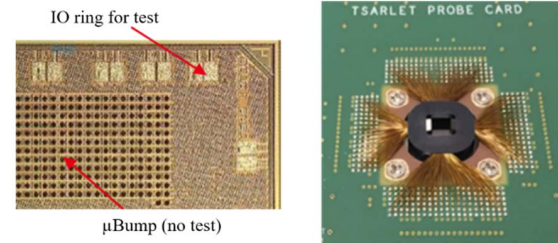


Figure 22. Chiplet test access and associated EWS probe card.



Figure 23. INTACT package and corresponding socket probe card.

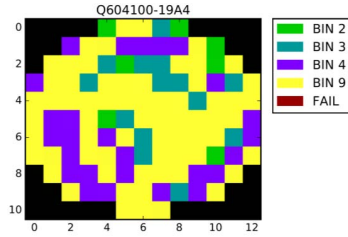


Figure 24. Chiplet EWS binning results.

V. CONCLUSION

Breaking with the System-on-Chip paradigm, future systems will consist in smaller, less expensive and independently designed components called chiplets, connected by high bandwidth interconnections over silicon interposers. We have explained the benefits of stacking chiplets on an active interposer for achieving large and scalable many-core systems for high performance computing. The INTACT prototype, embedding 96 cores within six chiplets, was introduced. We have detailed the full fabrication technology to enable turning this concept into reality: 3D interconnects process including TSV and ultra-fine pitch μ -pillars and the 3D packaging flow. The entire technology was assessed morphologically and electrically with specific 3D test structures. Finally, the structural test of the INTACT prototype was reached and is starting to show results well within expectations. The full extraction of the system performance associated to the architecture details is still ongoing. These results pave the way to the design of future high efficiency systems for high performance computing.

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